

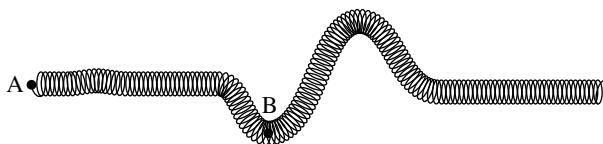
Chapter 10 The nature of waves

Section 10.1 Longitudinal and transverse waves

Section 10.1 Review

KEY QUESTIONS SOLUTIONS

- The particles oscillate back and forth or up and down around a central or average position and pass on the energy carried by the wave. They do not move along with the wave.
- False: Longitudinal waves occur when particles vibrate backwards and forwards about a mean position, parallel to the direction of the wave.
 - True.
 - True.
 - True.
- The pulse will move to the right and point B will move downwards.



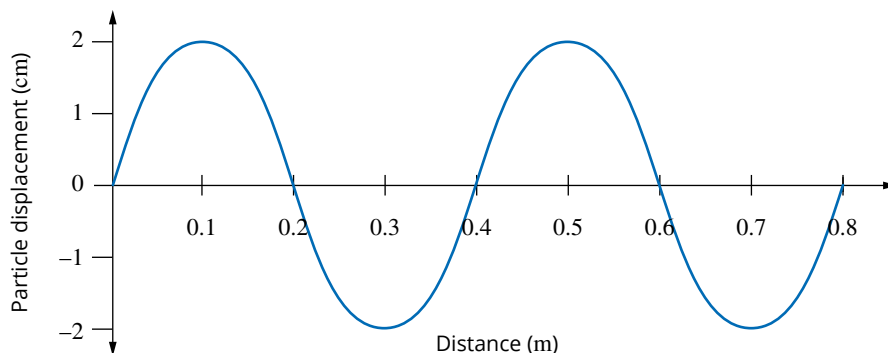
- Mechanical waves: sound, ripples on a pond, vibrations in a rope. (Light does not require the particles of a medium to propagate and is therefore not a mechanical wave.)
- The tuning fork vibrates back and forth, creating a series of compressions and rarefactions in the air as the energy is transferred.
- The forward motion of the source (for example, the speaker or tuning fork) pushes particles together so particle A goes to the right, the backward motion of the source creates a low-pressure area (the rarefaction) as particle B is moved to the left.
- In a transverse wave the motion of the particles is at right angles (perpendicular) to the direction of travel of the wave itself.
- Longitudinal: a and d
Transverse: b and c

Section 10.2 Representing waves

Worked example: Try yourself 10.2.1

DISPLACEMENT-DISTANCE GRAPHS 1

The displacement-distance graph below shows a snapshot of a transverse wave as it travels along a spring towards the right. Use the graph to determine the wavelength and the amplitude of this wave.

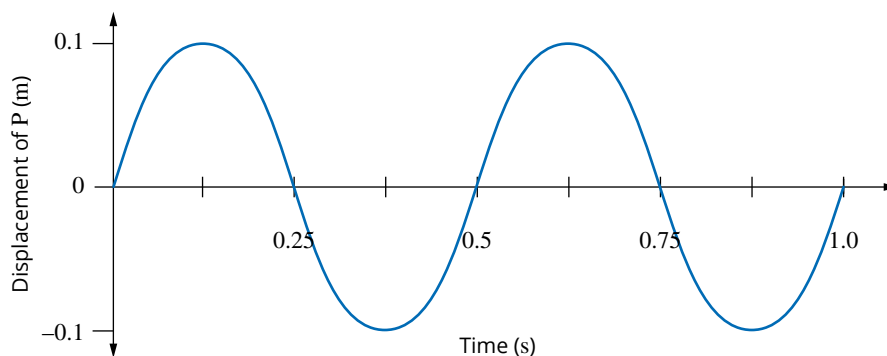


Thinking	Working
Amplitude on a displacement–distance graph is the distance from the average position to a crest or trough.	Amplitude = 2 cm = 0.02 m
Wavelength is the distance for one complete cycle. Any two consecutive points in phase and at the same position on the wave could be used.	Wavelength, $\lambda = 0.4$ m

Worked example: Try yourself 10.2.2

DISPLACEMENT–TIME GRAPHS 2

The displacement–time graph below shows the motion of a single part of a rope as a wave passes, travelling to the right. Use the graph to find the amplitude, period and frequency of the wave.



Thinking	Working
Amplitude on a displacement–time graph is the displacement from the average position to a crest or trough. Note the displacement of successive crests and/or troughs on the wave and carefully note units on the vertical axis.	Amplitude = 0.1 m
Period is the time it takes to complete one cycle and can be identified on a displacement–time graph as the time between two successive points that are in phase. Identify two points on the graph at the same position in the wave cycle. Confirm by checking two other points, e.g. two crests or two troughs.	Period, $T = 0.5$ s
Frequency can be calculated using $f = \frac{1}{T}$, measured in hertz (Hz).	$f = \frac{1}{T} = \frac{1}{0.5} = 2$ The frequency is 2 Hz.

Worked example: Try yourself 10.2.3

THE WAVE EQUATION 1

A longitudinal wave has a wavelength of 3.00 m and a speed of 1484 ms^{-1} . What is the frequency, f , of the wave?

Thinking	Working
The wave equation states that $v = f\lambda$. Knowing both v and λ , the frequency, f , can be found. Rewrite the wave equation in terms of f .	$v = f\lambda$ $f = \frac{v}{\lambda}$
Substitute the known values and solve.	$f = \frac{v}{\lambda}$ $= \frac{1484}{3.00}$ $= 495 \text{ Hz}$

Worked example: Try yourself 10.2.4

THE WAVE EQUATION 2

A longitudinal wave has a wavelength of 3.00 m and a speed of 1484 ms ⁻¹ . Calculate the period, T , of the wave.	
Thinking	Working
Rewrite the wave equation in terms of T .	$v = f\lambda, \text{ and } f = \frac{1}{T}$ $v = \frac{1}{T} \times \lambda$ $= \frac{\lambda}{T}$ $T = \frac{\lambda}{v}$
Substitute the known values and solve.	$T = \frac{\lambda}{v}$ $= \frac{3.00}{1484}$ $= 0.00202 \text{ s}$ $= 2.02 \times 10^{-3} \text{ s}$

Section 10.2 Review

KEY QUESTIONS SOLUTIONS

- C and F
 - wavelength
 - B and D
 - amplitude
- Wavelength is the length of one complete wave cycle. Any two points at the same position on the wave could be used. In this case $\lambda = 1.6 \text{ m}$.

Amplitude is the displacement from the average position to a crest or trough. In this case, amplitude = 20 cm.
- period = 0.4 s
 - $f = \frac{1}{T} = \frac{1}{0.4} = 2.5 \text{ Hz}$
- $f = 5 \text{ Hz}$, amplitude = 0.3 m, $\lambda = 1.3 \text{ m}$, $v = ?$

$v = f\lambda = 5 \times 1.3 = 6.5 \text{ ms}^{-1}$
- True.
 - False: The period of a wave is *proportional* to its wavelength.
 - True.
 - False: The wavelength *and frequency* of a wave determine its speed.
- wavelength = 4 cm; amplitude = 0.5 cm
 - $T = 2 \text{ s}$, $\lambda = 4 \text{ cm}$, $v = ?$
 $v = \frac{\lambda}{T} = \frac{4}{2} = 2 \text{ cm s}^{-1}$ or 0.02 ms^{-1}
 - red
- $T = \frac{1}{f} = \frac{1}{2 \times 10^5} = 5 \times 10^{-6} \text{ s}$
- As the speed of each vehicle is the same and there is no relative motion of the medium, the frequency observed would be the same as that at the source.
- The apparent frequency increases when travelling towards you, so the siren would sound higher in pitch, and decreases when travelling away from you, so the siren would sound lower.

Section 10.3 Wave behaviours—reflection, refraction and diffraction

Worked example: Try yourself 10.3.1

REFRACTION

Whales typically emit sounds between 10 and 40 Hz (humans can usually hear down to 20 Hz). If a whale emits a 20 Hz sound in water towards the surface at an angle of 40° to the normal, the refracted wave emerges from the water into air. The speed of sound in air is 343 ms^{-1} and the speed of sound in water is 1484 ms^{-1} .

a Find the wavelength in water and in air.	
Thinking	Working
The wavelength can be determined from $v = f\lambda$. Rearrange to make λ the subject: $\lambda = \frac{v}{f}$ where $f = 40 \text{ Hz}$ $v_{\text{air}} = 343 \text{ ms}^{-1}$ $v_{\text{water}} = 1484 \text{ ms}^{-1}$	$\lambda_{\text{water}} = \frac{1484}{40} = 37.1 \text{ m}$ $\lambda_{\text{air}} = \frac{343}{40} = 8.58 \text{ m}$
b Explain what will happen to the refracted wave, and why.	
The speed of sound in air is much slower than in water.	Since the speed in air is lower than in water, the angle will refract significantly towards the normal.
c Determine the angle of refraction.	
The angle of refraction can be calculated using Snell's law. $\frac{\sin r}{\sin i} = \frac{v_{\text{air}}}{v_{\text{water}}}$	$\sin r = \frac{v_{\text{air}}}{v_{\text{water}}} \sin i$ $\sin r = \frac{343}{1484} \sin 40^\circ$ $= 0.149$ $r = 8.6^\circ$

Worked example: Try yourself 10.3.2

TOTAL INTERNAL REFLECTION

Sound is travelling through air and hits a steel wall. At what angle is the sound totally reflected? The speed of sound in steel is 5000 ms^{-1} and the speed of sound in air is 340 ms^{-1} .	
Thinking	Working
The critical angle needs to be calculated using Snell's law $\frac{\sin r}{\sin i} = \frac{v_2}{v_1}$ Set $\sin r = 1$ $v_1 = 340 \text{ ms}^{-1}$ $v_2 = 5000 \text{ ms}^{-1}$	$\sin \theta_c = \frac{v_1}{v_2}$ $\sin \theta_c = \frac{340}{5000} = 0.068$ $\theta_c = 3.99^\circ$

Worked example: Try yourself 10.3.3

DIFFRACTION

In ultrasound imaging, the speed of sound is 1540 m s^{-1} . The resolution of an image depends on the wavelength of the sound—a smaller wavelength (higher frequency) enables more detail to be seen with less effect of diffraction. High-frequency sound (5 to 10 MHz) can resolve more detail but has limited penetration depth, whereas low-frequency sound (2 to 5 MHz) can penetrate to deeper structures but has lower resolution.

a If the human heart is 10 cm across, what frequency is needed to have at least 300 wavelengths across the image?	
Thinking	Working
The human heart is 10 cm across. 300 wavelengths need to fit into 10 cm.	$\lambda = \frac{10 \times 10^{-2}}{300}$ $= 0.000333 \text{ m}$
Calculate the frequency using: $f = \frac{v}{\lambda}$	$f = \frac{1540}{0.000333}$ $= 4.67 \times 10^6 \text{ Hz}$ $= 4.67 \text{ MHz}$
b If the frequency were significantly lower than your calculated amount, what would happen to the image? Explain why.	
As the frequency gets lower, the wavelength gets longer.	As the frequency gets lower, the image would be harder to resolve as diffraction effects would become greater.

Section 10.3 Review

KEY QUESTIONS SOLUTIONS

- The wave is reflected and there is a 180° change in phase.
- As a wave is reflected back into the same medium, the only property that will change is amplitude. This is because some of the energy of the wave has been absorbed by the second medium from which the wave was reflected. (Note: the velocity will change as the wave changes direction, but its speed will not change because it is a scalar quantity.)
- C. The object must be convex, that is, curved outwards.
- B. Angles are labelled relative to the normal.
- After an earthquake, P waves (longitudinal) can be detected after travelling through the region and can be refracted, but S waves (transverse) are not detected. P waves can travel through a fluid but S waves cannot. P waves are also refracted as they travel through the liquid interior and when they cross a boundary between different layers.
- B, C and D. Since refraction occurs travelling from deep water to shallow, it indicates there is a change in speed (D) which will cause a change in wavelength (B) and result in a change in angle or direction (C). The frequency of the wave (A) will remain unchanged.
- As wheel B rolls onto the grass it *slows down*. Since wheel A is now moving *faster* than wheel B, the wheels change *direction*. When wheel A rolls onto the grass the wheels' direction *stops* changing.
- $v_{20} = 331 + 0.60T = 331 + 0.60 \times 20 = 343 \text{ m s}^{-1}$
 - $v_{30} = 331 + 0.60T = 331 + 0.60 \times 30 = 349 \text{ m s}^{-1}$
 - The speed increases therefore the refracted angle will increase.
 - $\frac{\sin r}{\sin i} = \frac{v_{30}}{v_{20}}$ so $\sin r = \frac{v_{30} \times \sin i}{v_{20}} = \frac{349 \times \sin 50}{343} = 0.779$ and $r = 51.2^\circ$
 - Total internal reflection $\sin i = \frac{v_{20}}{v_{30}} = \frac{343}{349} = 0.983$, so $i = 79.4^\circ$. Therefore the angle must be greater than 79° .
- The higher frequency sound of the flute corresponds to a shorter wavelength so it will be diffracted less and will be more directional. Therefore, it will not be heard as well at the sides of the auditorium. The tuba undergoes a lot more diffraction and so will be louder at the sides.

Section 10.4 Wave interactions—superposition, interference and resonance

Section 10.4 Review

KEY QUESTIONS SOLUTIONS

- True.
 - False: As the pulses pass through each other, the interaction *does not* permanently alter the characteristics of each pulse.
 - True.
- B. Each pulse travels 3 m in 3 s. Adding their amplitudes together means they will look like C, but the result is they will cancel each other out as in B.
- An object subjected to forces with a forcing frequency matching its natural oscillating frequency will oscillate with increasing amplitude as there is a maximum transfer of energy. This could continue until the structure can no longer withstand the internal forces and fails.
- $\theta_i = 90^\circ - 38^\circ = 52^\circ$
 $\theta_r = \theta_i = 52^\circ$
- Normal walking results in a frequency of 1 Hz or 1 cycle per second i.e. two steps per second. This frequency may result in an increase in the amplitude of oscillation of the bridge over time, which could damage the structure.

Section 10.5 Standing waves and harmonics

Worked example: Try yourself 10.5.1

FUNDAMENTAL FREQUENCY

A standing wave in a string is found to have a wavelength of 0.50 m for the fundamental frequency of vibration. Assume that the tension in the string is not changed and that the string is fixed at both ends.

a Calculate the length of the string.	
Thinking	Working
Identify the wavelength of the string (λ) in metres and the harmonic number (n).	$\lambda = 0.5 \text{ m}$ $n = 1$
Recall that for any frequency, $\lambda = \frac{2\ell}{n}$. Rearrange to find ℓ .	$\lambda = \frac{2\ell}{n}$ $\ell = \frac{n\lambda}{2}$
Substitute the value for the wavelength from the question and solve for ℓ .	$\ell = \frac{1 \times 0.5}{2}$ $= 0.25 \text{ m}$

b Calculate the wavelength of the third harmonic.	
Thinking	Working
Identify length of the string (ℓ) in metres and the harmonic number (n)	$\ell = 0.25 \text{ m}$ $n = 3$
Recall that for any frequency $\lambda = \frac{2\ell}{n}$. Substitute the values from the question and solve for λ .	$\lambda = \frac{2\ell}{n}$ $= \frac{2 \times 0.25}{3}$ $= 0.17 \text{ m}$

Worked example: Try yourself 10.5.2
OPEN-ENDED AIR COLUMNS

The wavelength of the fourth harmonic in a tube that can be considered as an open-ended air column is found to be 12 cm.

a Calculate the length of the tube, assuming that the standing wave does not extend beyond the ends of the tube.	
Thinking	Working
Identify the wavelength of the sound (λ) in metres and the harmonic number (n).	$\lambda = 0.12 \text{ m}$ $n = 4$
Recall that for any frequency, $\lambda = \frac{2\ell}{n}$ Rearrange to find ℓ .	$\ell = \frac{n\lambda_n}{2}$ $= \frac{4 \times 0.12}{2}$ $= 0.24 \text{ m}$
b Determine the fundamental frequency.	
Thinking	Working
Recall for the fundamental frequency half a wavelength fits into the length of the pipe. $\ell = \frac{\lambda_1}{2}$	$\lambda_1 = 2\ell$ $= 2 \times 0.24$ $= 0.48 \text{ m}$
Then $f_1 = \frac{v}{\lambda_1}$	$f_1 = \frac{340}{0.48}$ $= 708 \text{ Hz}$

Worked example: Try yourself 10.5.3
AIR COLUMN CLOSED AT ONE END

An air column closed at one end is 12 cm long. Assume that the standing wave does not extend beyond the end of the tube.

a Calculate the wavelength of the fifth harmonic.	
Thinking	Working
Identify the length of the air column (ℓ) in metres and the harmonic number (n).	$\ell = 20 \text{ cm}$ $n = 5$
Recall that $\lambda_n = \frac{4\ell}{n}$ Substitute the values from the question and solve for λ .	$\lambda_5 = \frac{4\ell}{5}$ $= \frac{4 \times 0.20}{5}$ $= 0.16 \text{ m}$
b Calculate the frequency of the fifth harmonic if the velocity of sound is 340 ms^{-1} .	
Thinking	Working
Identify the speed of the sound (v) in ms^{-1} and the wavelength (λ) from the previous question.	$v = 340 \text{ ms}^{-1}$ $\lambda_5 = 0.16 \text{ m}$
Recall the wave equation $v = f\lambda$. Rearrange to find f .	$f_5 = \frac{v}{\lambda_5}$ $= \frac{340}{0.16}$ $= 2130 \text{ Hz}$

Section 10.5 Review

KEY QUESTIONS SOLUTIONS

- 1 It is a common misconception that standing waves somehow remain stationary. It is only the pattern made by the amplitude along the rope that stays still at the nodes. The rope is still moving, especially at the antinodes.
- 2 A transverse wave moving along a slinky spring is reflected at a fixed end with a phase change. The interference that occurs during the superposition of this reflected wave and the original wave creates a standing wave. This standing wave consists of locations called nodes, where the movement of the spring is cancelled out, and antinodes where maximum movement of the spring occurs. Nodes always occur at the ends.
- 3 $\lambda = \frac{2\ell}{n} = \frac{2 \times 0.4}{1} = 0.8 \text{ m}$
- 4 Rearranging $\lambda = \frac{2\ell}{n}$ gives $\ell = \frac{n\lambda}{2} = \frac{4 \times 0.75}{2} = 1.5 \text{ m}$
- 5 This wave will have a frequency four times that of the fundamental frequency, which means that it will have a wavelength $\frac{1}{4}$ of the fundamental wavelength due to the inverse relationship between frequency and wavelength.
- 6 The wavelength of the standing wave in the diagram is 5 m. The wavelength of the fundamental frequency is twice the length of the string. Therefore, a string length of 2.5 m would produce a standing wave with wavelength 5 m.
- 7 $f = \frac{nv}{2\ell}$
 $\ell = \frac{nv}{2f} = \frac{1 \times 387}{2 \times 350} = 0.55 \text{ m}$
 new length $= \frac{2}{3} \times 0.55 = 0.37 \text{ m}$
 new wavelength $= 2 \times \text{new length} = 2 \times 0.37 = 0.74 \text{ m}$
- 8 **a** $f = \frac{nv}{2\ell} = \frac{1 \times 300}{2 \times 0.5} = 300 \text{ Hz}$
b $f = \frac{nv}{2\ell} = \frac{2 \times 300}{2 \times 0.5} = 600 \text{ Hz}$
 Or use $f_2 = 2f_1 = 2 \times 300 = 600 \text{ Hz}$
c $f = \frac{nv}{2\ell} = \frac{3 \times 300}{2 \times 0.5} = 900 \text{ Hz}$
 Or use $f_3 = 3f_1 = 3 \times 300 = 900 \text{ Hz}$
- 9 Open-ended pipe, so use $\lambda_n = \frac{2\ell}{n}$
a $n = 1$, so $\lambda_1 = \frac{2\ell}{1} = \frac{2 \times 0.450}{1} = 0.900 \text{ m}$
b $n = 2$, so $\lambda_2 = \frac{2\ell}{2} = \frac{2 \times 0.450}{2} = 0.450 \text{ m}$
c $n = 3$, so $\lambda_2 = \frac{2\ell}{3} = \frac{2 \times 0.450}{3} = 0.300 \text{ m}$, and from $v = f\lambda$ rearrange:
 $f_3 = \frac{v}{\lambda_3} = \frac{330.0}{0.300} = 1100 \text{ Hz}$
- 10 The pipe is closed at one end, therefore use $f_n = \frac{nv}{4\ell}$
a Using $f_n = \frac{nv}{4\ell}$ where n is the harmonic number,
 $n = 1$ so $f_1 = \frac{1 \times v}{4\ell} = \frac{1 \times 330}{4 \times 0.750} = 110 \text{ Hz}$
b $n = 3$ so $f_3 = \frac{3 \times v}{4\ell} = \frac{3 \times 330}{4 \times 0.750} = 330 \text{ Hz}$
c Next harmonic along is $n = 5$ and $n = 7$
 So $f_5 = 5 \times f_1 = 550 \text{ Hz}$
 $f_7 = 7 \times f_1 = 770 \text{ Hz}$

- 11 a** $f_1 = 450 \text{ Hz}$. Each of the subsequent frequencies is double the previous, therefore all harmonics are formed and the pipe is open ended.
- b** $\lambda_1 = 3.0 \text{ m}$. The next resonance wavelength is $\frac{1}{3}$ that of the fundamental and the following is $\frac{1}{5}$ of the fundamental. This means the frequencies would be 3 times the fundamental and 5 times the fundamental, therefore the pipe is closed at one end.

Section 10.6 Wave intensity and applications of wave properties

Worked example: Try yourself 10.6.1

INTENSITY AND DISTANCE 1

Sam heard an annoying sound from 100 m away. By what factor would the intensity of the annoying sound change if Sam was to move to a distance of 400 m from the sound?	
Thinking	Working
Intensity, I , will decrease with the square of the distance, r , from the source. The ratio of the intensity at 400 m to the original intensity at 100 m is the factor required. Identify the initial values I_0 and r_0 and the final values I_f and r_f	$r_0 = 100 \text{ m}$ $I_0 \propto \frac{1}{r_0^2}$ then $I_0 \propto \frac{1}{100^2}$ $r_f = 400 \text{ m}$ $I_f \propto \frac{1}{r_f^2}$ then $I_f \propto \frac{1}{400^2}$
Determine the relationship between the intensity and radii.	$\frac{I_f}{I_0} = \frac{r_0^2}{r_f^2}$
Evaluate.	$\frac{I_f}{I_0} = \frac{100^2}{400^2}$ $\frac{I_f}{I_0} = 0.06$

Worked example: Try yourself 10.6.2

INTENSITY AND DISTANCE 2

A fog horn was originally heard from a boat when the boat was 1 km from the fog horn. After some time, the intensity of the fog horn was measured as being half of the original. Assuming the volume of the fog horn hadn't changed, how far away was the boat from the fog horn when the intensity was measured?	
Thinking	Working
Intensity, I , will decrease with the square of the distance, r , from the source. The expression $\frac{I_f}{I_0} = \frac{r_0^2}{r_f^2}$ can be used. Identify the variables r_0 and $\frac{I_f}{I_0}$.	$r_0 = 1000 \text{ m}$ $\frac{I_f}{I_0} = \frac{1}{2}$
Rearrange the expression and evaluate.	$\frac{I_f}{I_0} = \frac{r_0^2}{r_f^2}$ $r_f^2 = \frac{r_0^2 I_0}{I_f}$ $r_f^2 = \frac{1000^2}{\frac{1}{2}} = 1000^2 \times 2 = 2 \times 10^6$ $r_f = 141 \text{ m}$

Section 10.6 Review

KEY QUESTIONS SOLUTIONS

- $I_0 = 2.5 \times 10^4 \text{ W m}^{-2}$, $I_0 \propto \frac{1}{r_0^2} = \frac{1}{250^2}$ and $I_f \propto \frac{1}{r_f^2} = \frac{1}{1000^2}$, therefore $I_f = 2.5 \times 10^4 \times \frac{250^2}{1000^2} = 1560 \text{ W m}^{-2}$
- $r_0 = 20 \text{ m}$, $\frac{I_f}{I_0} = \frac{1}{8}$ then $\frac{I_f}{I_0} = \frac{r_0^2}{r_f^2}$ and $r_f^2 = \frac{r_0^2}{\frac{I_f}{I_0}} = \frac{20^2}{0.125} = 20^2 \times 8 = 3200$ therefore $r_f = 57 \text{ m}$
- At high-intensity the vibration of the tissues as the ultrasound is propagated produces heat, which can promote faster healing.
- Use an absorber like foam on the ceiling and carpets on the floor.
- A diffuser is a corrugated surface that allows reflections to occur in multiple directions, thus reducing echoes. Standing waves also cannot form.
- A high, solid barrier can be used to reflect the sound; vegetation can be used to absorb the sound.

CHAPTER 10 REVIEW

- The particles on the surface of the water move up and down as the waves radiate outwards, carrying energy away from the point on the surface of the water where the stone entered the water.
- Similarities: both are waves, both carry energy away from the source, both are caused by vibrations.
Differences: transverse waves involve particle displacement at right angles to the direction of travel of the wave; longitudinal waves involve particle displacement parallel to the direction of travel of the wave.
- U is moving down and V is momentarily stationary (and will then move downwards).
- $f = 10.0 \text{ Hz}$, $\lambda = 30.0 \text{ mm} = 0.0300 \text{ m}$, $v = ?$
 $v = f\lambda = 10 \times 0.03 = 0.300 \text{ m s}^{-1}$
- $f = 32\,000$, $v = 1400$, $\lambda = ?$
 $v = f\lambda$ rearranges to $\lambda = \frac{v}{f}$
 $\lambda = 1400 \div 32\,000 = 0.044 \text{ m}$
- C and D. Since the frequency rose and fell, the bike must have travelled past you. It must have come towards you and then moved away from you.
- By inspecting the wave equation $v = f\lambda$, since wavelength decreases and the velocity must stay the same, the frequency must increase. This ensures the product of the wavelength and frequency still equals the velocity, which has remained unchanged. (Note: velocity is constant as it is a property of the medium.)
- The green wave represents the superposition of the blue and the red waves as the amplitude of the green wave is the sum of the amplitudes of the blue and red waves.
- Sound waves are longitudinal mechanical waves where the particles only move back and forth around an equilibrium position, parallel to the direction of travel of the wave. When these particles move in the direction of the wave, they collide with adjacent particles and transfer energy to the particles in front of them. This means that kinetic energy is transferred between particles in the direction of the wave through collisions. Therefore, the particles cannot move along with the wave from the source as they lose their kinetic energy to the particles in front of them during the collisions.
- C and D. Only energy is transferred by a wave therefore the statements saying that air particles have travelled to Lee are incorrect. Energy has been transferred from the speaker to Lee and it is the air particles that have passed this energy along through the air.
- All objects/materials have a resonant frequency. If the object is made to vibrate at this frequency, the amplitude of the object's vibrations will increase with time. If a building or bridge was subjected to wind that made it vibrate at its natural frequency, this vibration may increase in amplitude so much that the structure is damaged or collapses.
- $f = \frac{nv}{4\ell} = \frac{1 \times 340}{4 \times 0.85} = 100 \text{ Hz}$
- $f = \frac{nv}{4\ell} = \frac{3 \times 340}{4 \times 0.85} = 300 \text{ Hz}$

- 14** The fundamental frequency is given by:

$$f_1 = \frac{1}{T} = \frac{1}{4.0} = 0.25 \text{ Hz}$$

The frequency of the second harmonic is given by:

$$f_2 = 2 \times f_1 = 2 \times 0.25 = 0.50 \text{ Hz}$$

- 15** Calculate the wavelength of the wave using the wave equation:

$$v = f\lambda$$

$$\lambda = \frac{v}{f}$$

$$= \frac{78}{428}$$

$$= 0.182 \text{ m}$$

Since the separation of antinodes and of nodes in a standing wave in a string with fixed ends is half the wavelength, then:

$$d = \frac{\lambda}{2} = \frac{0.182}{2}$$

$$= 0.091 \text{ m or } 9.1 \text{ cm}$$

- 16** All of the options are correct. The light rays striking all of these surfaces will obey the law of reflection, as it always holds regardless of the shape of the reflector.

- 17** B. In resonance, maximum energy is transferred and the amplitude of vibration will increase. The frequency is unchanged.

18 $\lambda_3 = \frac{v}{f_3} = \frac{330}{400} = 0.825 \text{ m}; \lambda_3 = \frac{2\ell}{n} \text{ so } \ell = \frac{n \times \lambda_3}{2} = \frac{3 \times 0.825}{2} = 1.24 \text{ m}$

19 a The harmonic $n = 1$: $f_1 = \frac{1 \times v}{4\ell} = \frac{1 \times 340}{4 \times 0.85} = 100 \text{ Hz}$

b Third harmonic: $f_3 = \frac{3 \times v}{4\ell} = \frac{3 \times 340}{4 \times 0.85} = 300 \text{ Hz, or } f_3 = 3 \times f_1 = 300 \text{ Hz}$

- 20** The angle of refraction from the normal to the refracted ray would decrease relative to the angle of incidence. The speed of sound in air is less than the speed of sound in water, therefore the refracted angle would be smaller.

- 21** Total internal reflection occurs when the wave goes from a slow-speed medium to a higher-speed medium. The refracted angle increases. At the critical angle the angle of reflection is at 90° and lies along the interface between the two media. Any angles greater than this will reflect back into the first medium.

22 D. $I_1 = x$, $r_2 = 2r_1$, $I_1 \propto \frac{1}{r_1^2}$ and $I_2 \propto \frac{1}{r_2^2} = \frac{1}{(2r_1)^2} = \frac{1}{4r_1^2}$ therefore $I_2 = \frac{1}{4}I_1 = \frac{x}{4}$

- 23** Increasing the distance by a factor of 4, from 1000 m to 4000 m, reduces the intensity by a factor of 16, from 1 W m^{-2} to $\frac{1}{16} \text{ W m}^{-2}$. Therefore, to collect 1 W the dish would need to have an area of 16 m^2 . Using $A = \pi r^2$,

then $r = \sqrt{\frac{A}{\pi}} = \sqrt{\frac{16}{\pi}} = 2.26 \text{ m}$, therefore the diameter is 4.514 m (to the nearest mm).